

- 5 (a) Define electric potential at a point.

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.....

..... [2]

- (b) A hydrogen atom may be considered to consist of a proton and an electron separated by a distance of 120 pm, as shown in Fig. 5.1.

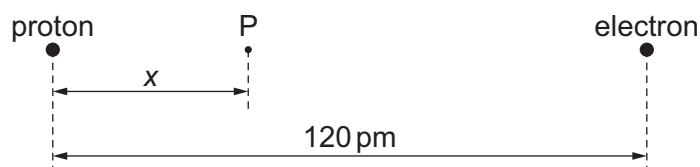


Fig. 5.1

The two particles may be considered as point charges.

Point P lies on the line joining the electron and the proton and is at a variable distance x from the proton.

- (i) Show that the electric potential V at point P when $x = 10$ pm is equal to 130 V.

[2]

- (ii) Calculate, to two significant figures, V when $x = 30$ pm.

$V = \dots\dots\dots V$ [2]

- (iii) On Fig. 5.1, draw a cross (x) at one position, other than infinity, where the electric potential is zero. [1]

